Homework 4 Solutions

Part I: Key Concepts

(**1**) Answer each of the following questions in a few sentences.

(a) Why do we use the pooled sample proportion $\hat{p} = \frac{x_1+x_2}{n_1+n_2}$ in the standard error SE_0 when we test $H_0 : p_1 = p_2$, and what is the benefit of pooling?

A test works out what the data should look like **if H_0 is true**. If $p_1 = p_2$, both samples describe the **same** population proportion p, so the best estimate of that one p combines all of the data: all the successes over all $n_1 + n_2$ observations. The standard error is then built from what H_0 says, $SE_0 = \sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$, just as the one-sample test uses p_0 in its SE. The benefit is that $\hat{p}$ uses all $n_1 + n_2$ observations instead of n_1 or n_2 alone, so it is a more precise estimate of p, and SE_0 is the standard error that matches the null distribution of $\hat{p}_1 - \hat{p}_2$, which keeps the Type I error rate at α.

(b) What are the assumptions of the test comparing two population proportions?

(1) Two **independent random samples**, one from each population (or two groups formed by random assignment). (2) **Large samples**: with the pooled $\hat{p}$, all four of $n_1\hat{p}$, $n_1(1-\hat{p})$, $n_2\hat{p}$ and $n_2(1-\hat{p})$ are at least 15, so the sampling distribution of $\hat{p}_1 - \hat{p}_2$ is approximately normal.

(c) What does the pooled standard deviation $s_{pooled} = \sqrt{\frac{(n_1-1)s_1^2+(n_2-1)s_2^2}{n_1+n_2-2}}$ estimate in a two-sample t-test, and when is it appropriate to use it?

If the two populations have the **same standard deviation**, $\sigma_1 = \sigma_2 = \sigma$, both samples estimate the same σ, and s_{pooled} combines them into one estimate of it (a weighted average of the two sample variances, weighted by their degrees of freedom). It is appropriate only when we can assume $\sigma_1 = \sigma_2$: the sample SDs are similar (the larger is less than about twice the smaller) and the sample sizes are similar, or we are told to assume equal variances. Unlike the test of two proportions, $H_0 : \mu_1 = \mu_2$ says nothing about σ_1 and σ_2, so the unpooled test is our default.

(d) Give the formula for Welch's degrees of freedom for the unpooled two-sample t-test. How does it differ from using $df = \min(n_1 - 1, n_2 - 1)$?

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}$$

It is the df that makes a t curve match the distribution of t_{obs} most closely. It is usually not a whole number and always lies between $\min(n_1 - 1, n_2 - 1)$ and $n_1 + n_2 - 2$; software uses it. $\min(n_1 - 1, n_2 - 1)$ is a quick by-hand approximation that is **never larger** than the Welch df, so its t curve has heavier tails and its p-value comes out a little larger. That makes the hand method **conservative**: if it rejects H_0, the Welch test will too.

(e) Fill in the blanks. When we can assume $\sigma_1 = \sigma_2$, the pooled two-sample t-test has _____ degrees of freedom than the unpooled test, so its t-distribution has _____ tails and the test has _____ power.

More degrees of freedom ($n_1 + n_2 - 2$, against $\min(n_1 - 1, n_2 - 1)$ by hand or the Welch df in software), so **thinner** tails, smaller p-values and **more** power. That is the payoff of pooling, but only when the equal-SD assumption is true; if the SDs differ, the pooled test's real Type I error rate can be far from α.

(f) What are the assumptions of the two independent samples t-test? What extra assumption does the pooled version need?

(1) Two **independent random samples**, one from each population (or two groups formed by random assignment), with nothing in one sample paired with anything in the other. (2) **Normal populations or large samples**: if both n_1 and n_2 are at least 30, the Central Limit Theorem takes care of the shape; with smaller samples, both populations should be approximately normal (no strong skew or outliers). The pooled test also assumes the two populations have the **same standard deviation**, $\sigma_1 = \sigma_2$.

(g) When is it appropriate to use a paired t-test, and how does it differ from a two independent samples t-test?

A paired t-test is appropriate when each observation in one sample is **matched** to one observation in the other: both are measured on the same individual (before and after a treatment, or under two conditions) or on a natural pair (twins, the two hands of one person). For each pair we compute the difference d, always in the same order, and run a **one-sample** t-test on the differences, with parameter μ_d, $SE = s_d/\sqrt{n}$ and $df = n - 1$, where n is the number of **pairs**. A two independent samples t-test compares two separate groups with nothing linking them, with parameter $\mu_1 - \mu_2$ and $SE = \sqrt{s_1^2/n_1 + s_2^2/n_2}$. Pairing removes the person-to-person variation, so s_d is usually much smaller than s_1 and s_2, which gives a smaller SE and a more powerful test. Analyzing paired data as two independent samples is wrong, because the two samples are not independent.

(h) What are the assumptions of the paired t-test?

(1) The pairs are a **random sample** (or the treatment was randomly assigned within each pair), and the pairs are independent of one another. (2) The **differences** are approximately normal, or there are at least 30 pairs. The two original measurements do not each need to be normal; only the differences matter.

(i) What are the observed counts and the expected counts in a chi-square test, and how is each expected count computed?

The **observed** counts are the numbers of individuals in each category (or cell) in the sample. The **expected** counts are the numbers we would expect in each category if the null hypothesis were true: $n \times p_i$ for a goodness-of-fit test, and $\frac{\text{row total} \times \text{column total}}{\text{grand total}}$ for a test of independence.

(j) What are the parameters of a chi-square distribution? What are their values for a goodness-of-fit test with k categories, and for a test of independence on a table with r rows and c columns?

A chi-square distribution has a **single parameter, its degrees of freedom** df, which sets its shape (its mean is df). The distribution starts at 0 and is right-skewed, and it becomes more spread out and more symmetric as df grows. Goodness of fit: $df = k - 1$. Test of independence: $df = (r - 1)(c - 1)$.

(k) What is the difference between a goodness-of-fit test and a test of independence, and when is each test appropriate?

A **goodness-of-fit** test uses **one** categorical variable and asks whether its distribution in the population matches a claimed set of probabilities (for example, a fair die, or a known population distribution); its hypotheses are about the category probabilities $p_1, \ldots, p_k$. A **test of independence** uses **two** categorical variables measured on the same individuals, arranged in a two-way table, and asks whether they are related; H_0 says the two variables are independent. Use a goodness-of-fit test to compare one variable with a claimed distribution, and a test of independence to ask whether two variables are associated.

(l) What are the assumptions of the chi-square goodness-of-fit test and the chi-square test of independence?

Both tests assume (1) the data are a **random sample**, with each individual counted in exactly one category (or cell), and (2) **every expected count is at least** 5, so the chi-square distribution is a good approximation to the distribution of χ^2_{obs}. For the test of independence, this applies to every one of the $r \times c$ expected cell counts.

(m) Why is a chi-square test always an upper-tailed test?

$\chi^2_{obs} = \sum \frac{(O-E)^2}{E}$ adds up squared distances, so it is never negative and it is close to 0 when the observed counts match the expected counts. Only a **large** value is evidence against H_0, so the p-value is the area in the upper tail, $P(\chi^2 \geq \chi^2_{obs} \mid H_0 \text{ true})$.

Part II: Choosing a Test

(**2**) For each study below, (i) decide whether the samples are **independent** or **dependent (paired)** and explain why, and (ii) name the most appropriate test: the test comparing two proportions, the two independent samples t-test, the paired t-test, the chi-square goodness-of-fit test, or the chi-square test of independence. You do not need to carry out the test.

(a) A running-shoe company claims that its new carbon-plated racing shoe makes runners faster. A coach recruits 20 members of a university cross-country team. Each runner runs a 5 km time trial on the same course twice, one week apart: once in their usual training shoes and once in the new shoes, with a coin flip deciding which pair they wear first. The coach records each runner's finishing time in minutes and wants to know whether times are lower, on average, in the new shoes.

(i) **Dependent.** Each runner is measured twice, so every time in the new shoes is paired with the same runner's time in the usual shoes. A fast runner is fast in either pair of shoes, so the two times from one runner are related. (ii) **Paired t-test** on the 20 differences $d = \text{new} - \text{usual}$, with $H_0 : \mu_d = 0$ against $H_A : \mu_d < 0$ (lower-tailed) and $df = 20 - 1 = 19$.

(b) A university's advising office believes that text-message reminders get more first-year students to use its free tutoring center. At the start of the fall semester, 1,200 first-year students are randomly split into two groups of 600: one group receives a weekly text reminder about the tutoring center and the other receives none. At the end of the semester, the office records whether each student visited the tutoring center at least once.

(i) **Independent.** The two groups are different students, formed by random assignment, and no student in one group is matched with any student in the other. (ii) The response (visited or not) is categorical with two outcomes, so this is the **test comparing two proportions**: p_1 and p_2 are the proportions of students with and without reminders who visit the center, $H_0 : p_1 = p_2$ against $H_A : p_1 > p_2$ (upper-tailed), using the pooled $\hat{p}$ in SE_0.

(c) An animal scientist suspects that adding a seaweed supplement to dairy cattle feed changes how much milk the cows produce, but she does not know in which direction. Of the 80 cows in a university research herd, 40 are randomly assigned to the standard diet and 40 to the standard diet plus the supplement. After eight weeks, she records each cow's average milk yield in kilograms per day.

(i) **Independent.** Each cow is in only one group, the groups were formed by random assignment, and nothing links a cow in one group to a particular cow in the other. (ii) The response (milk yield) is quantitative, so this is the **two independent samples t-test**: $H_0 : \mu_1 - \mu_2 = 0$ against $H_A : \mu_1 - \mu_2 \neq 0$ (two-sided, since the direction is unknown), unpooled by default with $df = \min(40 - 1, 40 - 1) = 39$.

(d) A hospital's physical therapy team thinks a new rehabilitation program shortens recovery after knee-replacement surgery. They recruit 36 patients and form 18 pairs, matching the two patients in each pair on age, sex and body mass index. Within each pair, a coin flip sends one patient to the new program and the other to the standard program, and the team records the number of days until each patient can walk unassisted.

(i) **Dependent.** The two samples are different people, but each patient in the new program is deliberately matched to one patient in the standard program, on traits that affect how fast people recover. The pairs are what make the samples dependent, not whether the same person is measured twice. (ii) **Paired t-test** on the 18 within-pair differences d = new − standard, with $H_0 : \mu_d = 0$ against $H_A : \mu_d < 0$ (lower-tailed) and $df = 18 - 1 = 17$. Treating the data as two independent samples of 18 would ignore the matching and give a larger SE.

Part III: Comparing Two Population Proportions p_1 and p_2

(**3**) Wearable fitness devices: $x_1 = 379$ of $n_1 = 973$ people in their twenties, $x_2 = 404$ of $n_2 = 1{,}304$ people in their thirties, $\alpha = 0.05$, upper-tailed.

```
two.sample.prop.test(p0 = 0, x1 = 379, x2 = 404, n1 = 973, n2 = 1304,
                     alpha = 0.05, test = 'upper.tail', pooled = TRUE)
```

```
## [1] ==================== test results ====================
## [1] test type = upper.tail
## [1] H0: p1 - p2 = 0
## [1] HA: p1 - p2 > 0
## [1] estimate = 0.0797
## [1] pooled SE = 0.0201
## [1] 95% CI = [0.0465,0.1129]
## [1] Critical Value = 1.6449
## [1] Test statistic = 3.9608
## [1] Pvalue = 0
## [1] Decision: reject H0
## [1] ======================================================
```

(a) Check the assumptions of the test.

The two samples are independent random samples. The pooled proportion is

$$\hat{p} = \frac{379 + 404}{973 + 1{,}304} = \frac{783}{2{,}277} = 0.3439$$

and the four counts are $n_1\hat{p} = 334.6$, $n_1(1-\hat{p}) = 638.4$, $n_2\hat{p} = 448.4$ and $n_2(1-\hat{p}) = 855.6$, all at least 15. The large-sample assumption holds.

(b) Define p_1 and p_2 in words and state the null and alternative hypotheses.

p_1 is the proportion of all people aged 21–30 who use wearable fitness devices, and p_2 is the proportion of all people aged 31–40 who do. $H_0 : p_1 - p_2 = 0$ and $H_A : p_1 - p_2 > 0$ (upper-tailed).

(c) Compute $\hat{p}_1$, $\hat{p}_2$, the standard error under H_0, and the test statistic z_{obs}.

$\hat{p}_1 = 379/973 = 0.3895$ and $\hat{p}_2 = 404/1{,}304 = 0.3098$.

$$SE_0 = \sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)} = \sqrt{0.3439(0.6561)\left(\frac{1}{973} + \frac{1}{1{,}304}\right)} = 0.0201$$

$$z_{obs} = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{SE_0} = \frac{0.3895 - 0.3098}{0.0201} \approx 3.96$$

(d) Compute the p-value.

$$P(Z \geq 3.96 \mid H_0 \text{ true}) < 0.0002$$

3.96 is beyond the end of the Z-table ($P(Z \leq 3.49) = 0.9998$), so the p-value is less than 0.0002. The exact value is about 0.00004.

(e) Is there enough evidence to conclude that a higher proportion of people in their twenties use wearable fitness devices than people in their thirties?

Yes. The p-value $< 0.0002 < \alpha = 0.05$ (equivalently, $z_{obs} = 3.96 > z_{0.95} = 1.645$), so we reject H_0. At the 5% significance level there is strong evidence that a higher proportion of people in their twenties use wearable fitness devices than people in their thirties (about 39% vs 31% in the samples).

(**4**) COVID-19 and child-abuse registries: $x_1 = 18$ of $n_1 = 70$ registries before COVID-19, $x_2 = 16$ of $n_2 = 53$ after, $\alpha = 0.01$, two-sided.

```
two.sample.prop.test(p0 = 0, x1 = 18, x2 = 16, n1 = 70, n2 = 53,
                     alpha = 0.01, test = 'two.tail', pooled = TRUE)
```

```
## [1] ==================== test results ====================
## [1] test type = two.tail
## [1] H0: p1 - p2 = 0
## [1] HA: p1 - p2 != 0
## [1] estimate = -0.0447
## [1] pooled SE = 0.0814
## [1] 99% CI = [0,0.1662]
## [1] Critical Value = 2.5758
## [1] Test statistic = -0.5495
## [1] Pvalue = 0.5827
## [1] Decision: fail to reject H0
## [1] ======================================================
```

(a) Compute the pooled sample proportion $\hat{p}$ and check the four large-sample counts.

$$\hat{p} = \frac{18 + 16}{70 + 53} = \frac{34}{123} = 0.2764$$

The counts are $n_1\hat{p} = 19.35$, $n_1(1 - \hat{p}) = 50.65$, $n_2\hat{p} = 14.65$ and $n_2(1 - \hat{p}) = 38.35$. One count, $n_2\hat{p} = 14.65$, is just under 15, so the large-sample assumption is not quite met. The sampling distribution of $\hat{p}_1 - \hat{p}_2$ may not be close enough to normal for the standard normal p-value to be accurate, so the p-value is only an approximation and we should be cautious with a decision near the cut-off. (The shortfall here is small, and the textbook's looser rule of at least 5 successes and 5 failures is met.)

(b) Carry out the remaining four steps of the test.

Hypotheses. Let p_1 be the proportion of child-abuse registries in Sergipe involving a child under 12 before COVID-19 (2019), and p_2 the same proportion after COVID-19 started (2020). $H_0 : p_1 - p_2 = 0$ and $H_A : p_1 - p_2 \neq 0$.

Test statistic. $\hat{p}_1 = 18/70 = 0.2571$ and $\hat{p}_2 = 16/53 = 0.3019$.

$$SE_0 = \sqrt{0.2764(0.7236)\left(\frac{1}{70} + \frac{1}{53}\right)} = 0.0814, \qquad z_{obs} = \frac{0.2571 - 0.3019}{0.0814} \approx -0.55$$

p-value.

$$2 \times P(Z \geq |-0.55| \mid H_0 \text{ true}) = 2 \times 0.2912 = 0.5824$$

Decision. The p-value $0.5824 > \alpha = 0.01$ (equivalently, $|z_{obs}| = 0.55 < z_{0.995} = 2.576$), so we fail to reject H_0. At the 1% significance level there is not enough evidence that the proportion of abuse registries involving children under 12 changed after the start of the COVID-19 pandemic. The test statistic is so far from the cut-off that the concern from part (a) does not change this conclusion.

Part IV: Comparing Two Independent Population Means μ_1 and μ_2

(**5**) Fitness as a motivating factor: team sports $\bar{x}_1 = 7.79$, $s_1 = 1.27$, $n_1 = 203$; individual sports $\bar{x}_2 = 8.40$, $s_2 = 0.94$, $n_2 = 62$; pooled, $\alpha = 0.05$, two-sided.

```
two.sample.t.test(m0 = 0, x1 = 7.79, x2 = 8.40, s1 = 1.27, s2 = 0.94, n1 = 203, n2 = 62,
                  alpha = 0.05, test = 'two.tail', pooling = 'pooled')
```

```
## [1] ==================== test results ====================
## [1] test type = two.tail
## [1] H0: m1 - m2 = 0
## [1] HA: m1 - m2 != 0
## [1] estimate = -0.61
## [1] pooled SE = 0.1744
## [1] 95% CI = [-0.9034,-0.3166]
## [1] Critical Value = 1.969
## [1] Test statistic = -3.4987
## [1] degrees of freedom = 263
## [1] Pvalue = 5e-04
## [1] Decision: reject H0
## [1] ========================================================
```

Assumptions. The two samples are independent random samples (stratified random sampling of athletes), both samples are large ($n_1 = 203$ and $n_2 = 62$ are well over 30), and we are told to assume the two populations have the same variance.

Hypotheses. Let μ_1 be the mean fitness motivation score of all team-sport athletes and μ_2 the mean fitness motivation score of all individual-sport athletes. $H_0 : \mu_1 - \mu_2 = 0$ and $H_A : \mu_1 - \mu_2 \neq 0$.

Test statistic.

$$s_{pooled} = \sqrt{\frac{(203-1)(1.27)^2 + (62-1)(0.94)^2}{203+62-2}} = 1.202$$

$$t_{obs} = \frac{(7.79-8.40)-0}{1.202\sqrt{\frac{1}{203}+\frac{1}{62}}} = \frac{-0.61}{0.1744} \approx -3.50, \qquad df = 203+62-2 = 263$$

p-value. With 263 degrees of freedom the t-distribution is essentially the standard normal, so use the Z-score row of the t-table. $|t_{obs}| = 3.50$ is larger than 3.291, the two-tailed value for 0.001, so

$$2 \times P(t \geq 3.50 \mid H_0 \text{ true}) < 0.001$$

(the exact value is about 0.0005).

Decision. The p-value $< 0.001 < \alpha = 0.05$ (equivalently, $|t_{obs}| = 3.50 > 1.96$), so we reject H_0. At the 5% significance level the mean fitness motivation score is significantly different for team and individual sports; individual-sport athletes rate fitness as a more important reason to participate (8.40 vs 7.79).

(**6**) Four-year-old male frogs: Servotte $\bar{x}_1 = 64.7$, $s_1 = 7.3$, $n_1 = 12$; Thevenon $\bar{x}_2 = 61.0$, $s_2 = 4.1$, $n_2 = 12$; unpooled, $\alpha = 0.05$, two-sided.

```
two.sample.t.test(m0 = 0, x1 = 64.7, x2 = 61.0, s1 = 7.3, s2 = 4.1, n1 = 12, n2 = 12,
                  alpha = 0.05, test = 'two.tail', pooling = 'approx.unpooled')
```

```
## [1] ==================== test results ====================
## [1] test type = two.tail
## [1] H0: m1 - m2 = 0
## [1] HA: m1 - m2 != 0
## [1] estimate = 3.7
## [1] approx.unpooled SE = 2.417
```

```
## [1] 95% CI = [-1.6197,9.0197]
## [1] Critical Value = 2.201
## [1] Test statistic = 1.5309
## [1] degrees of freedom = 11
## [1] Pvalue = 0.154
## [1] Decision: fail to reject H0
## [1] ======================================================
```

Assumptions. The two samples are independent random samples from the two populations. Both samples are small ($n = 12$), so we must assume body lengths are approximately normal in each population. The variances are not assumed equal, so we use the unpooled standard error.

Hypotheses. Let μ_1 be the mean body length of four-year-old male frogs in the Servotte population and μ_2 the mean body length of four-year-old male frogs in the Thevenon population. $H_0 : \mu_1 - \mu_2 = 0$ and $H_A : \mu_1 - \mu_2 \neq 0$.

Test statistic.

$$t_{obs} = \frac{(64.7 - 61.0) - 0}{\sqrt{\frac{7.3^2}{12} + \frac{4.1^2}{12}}} = \frac{3.7}{2.417} \approx 1.53, \qquad df = \min(12 - 1, 12 - 1) = 11$$

p-value. In the 11 df row of the t-table, 1.53 falls between 1.363 (two-tailed 0.2) and 1.796 (two-tailed 0.1), so

$$0.1 < 2 \times P(t \geq 1.53 \mid H_0 \text{ true}) < 0.2$$

(the exact value is about 0.154).

Decision. The p-value $> \alpha = 0.05$ (equivalently, $|t_{obs}| = 1.53 < t_{11,0.975} = 2.201$), so we fail to reject H_0. At the 5% significance level there is not enough evidence that the mean body length of four-year-old male frogs differs between the Servotte and Thevenon populations.

Part V: Paired Samples

(**7**) Cholesterol and a low-fat diet: 10 people, d = Ending − Starting, $\alpha = 0.05$.

Table 1: Cholesterol before and after the diet, with the differences filled in

Person	Starting Cholesterol (mg/dL)	Ending Cholesterol (mg/dL)	Difference: d = Ending - Starting
1	140	140	0
2	220	230	10
3	110	120	10
4	240	220	-20
5	200	190	-10
6	180	150	-30
7	190	200	10
8	360	300	-60
9	280	300	20
10	260	240	-20

```
d = end - start
one.sample.t.test(m0 = 0, xbar = mean(d), s = sd(d), n = length(d),
                  alpha = 0.05, test = 'lower.tail', verbose = T)
```

```
## [1] ==================== test results ====================
## [1] test type = lower.tail
## [1] H0: m0 = 0
## [1] HA: m < 0
## [1] Estimate = -9
## [1] Estimated Standard Error = 7.6667
## [1] Critical Value = -1.8331
## [1] 95% CI = [0,8.3432]
## [1] Test statistic = -1.1739
## [1] Degrees of Freedom = 9
## [1] Pvalue = 0.1353
## [1] Decision: fail to reject H0
## [1] ======================================================
```

(a) Explain why these are paired (dependent) samples and not two independent samples.

Both measurements come from the same ten people: each starting level is matched to the ending level of the same person. The two columns are not separate samples from two populations, so we analyze the ten within-person differences.

(b) Fill in the difference column and compute $\bar{d}$ and s_d.

The differences are shown in the table above. $\bar{d} = -90/10 = -9$ mg/dL and $s_d = 24.24$ mg/dL.

(c) Define μ_d in words and state the null and alternative hypotheses.

μ_d is the mean change in cholesterol (ending minus starting) for all people who follow the 12-week low-fat diet. A drop in cholesterol gives a negative difference, so $H_0 : \mu_d = 0$ and $H_A : \mu_d < 0$ (lower-tailed).

(d) Compute the test statistic t_{obs} and give its degrees of freedom.

$$t_{obs} = \frac{\bar{d} - 0}{s_d/\sqrt{n}} = \frac{-9}{24.24/\sqrt{10}} = \frac{-9}{7.67} \approx -1.17, \qquad df = 10 - 1 = 9$$

(e) Compute the p-value.

By symmetry, $P(t \leq -1.17) = P(t \geq 1.17)$. In the 9 df row, 1.17 falls between 1.100 (upper tail 0.15) and 1.230 (upper tail 0.125), so

$$0.125 < P(t \leq -1.17 \mid H_0 \text{ true}) < 0.15$$

(the exact value is about 0.135).

(f) Is there enough evidence to conclude that the low-fat diet lowered cholesterol levels?

No. The p-value $> \alpha = 0.05$ (equivalently, $t_{obs} = -1.17 > t_{9,0.05} = -1.833$), so we fail to reject H_0. Although cholesterol fell by 9 mg/dL on average, at the 5% significance level there is not enough evidence that the diet lowers mean cholesterol: the changes vary too much from person to person (from a rise of 20 to a drop of 60) for ten people to rule out ordinary variation.

(**8**) Breakfast and work hours: 10 pairs, $d = \text{With} - \text{Without}$, $\alpha = 0.05$.

Table 2: Hours worked, with the differences filled in

Pair	Work Hours with Breakfast	Work Hours without Breakfast	Difference: d = With - Without
1	8	6	2
2	7	5	2
3	9	5	4
4	5	4	1
5	9	7	2
6	8	7	1
7	10	7	3
8	7	5	2
9	6	6	0
10	9	5	4

```
d = with_b - without_b
one.sample.t.test(m0 = 0, xbar = mean(d), s = sd(d), n = length(d),
                  alpha = 0.05, test = 'upper.tail', verbose = T)
```

```
## [1] ==================== test results ====================
## [1] test type = upper.tail
## [1] H0: m0 = 0
## [1] HA: m > 0
## [1] Estimate = 2.1
## [1] Estimated Standard Error = 0.4069
## [1] Critical Value = 1.8331
## [1] 95% CI = [1.1796,3.0204]
## [1] Test statistic = 5.1612
## [1] Degrees of Freedom = 9
## [1] Pvalue = 3e-04
## [1] Decision: reject H0
## [1] ======================================================
```

Assumptions. The data are paired: each person worked once without breakfast and once with it. With only 10 pairs, we must assume the differences are approximately normal. A positive difference means the person worked longer on the day they ate breakfast.

Hypotheses. Let μ_d be the mean difference in hours worked (with breakfast minus without). $H_0 : \mu_d = 0$ and $H_A : \mu_d > 0$ (upper-tailed).

Test statistic. The differences are $2, 2, 4, 1, 2, 1, 3, 2, 0, 4$, so $\bar{d} = 2.1$ hours and $s_d = 1.287$ hours.

$$t_{obs} = \frac{2.1 - 0}{1.287/\sqrt{10}} = \frac{2.1}{0.407} \approx 5.16, \qquad df = 9$$

p-value. In the 9 df row, 5.16 is beyond 4.781 (upper tail 0.0005), so

$$P(t \geq 5.16 \mid H_0 \text{ true}) < 0.0005$$

(the exact value is about 0.0003).

Decision. The p-value $< 0.0005 < \alpha = 0.05$ (equivalently, $t_{obs} = 5.16 > t_{9,0.95} = 1.833$), so we reject H_0. At the 5% significance level there is strong evidence that eating breakfast increases the mean number of hours worked; in this sample, people worked about 2.1 hours longer after breakfast.

Part VI: Chi-Square Goodness-of-Fit Tests

(**9**) A die rolled 120 times, $\alpha = 0.05$.

```
chi.squared.GOF.test(observed.ct = c(15, 29, 16, 15, 30, 15), expected.freq = rep(1/6, 6), alpha = 0.05)
```

```
## [1] ==================== test results ====================
## [1] H0: P(category 1) = 0.1667 H0: P(category 2) = 0.1667
## [3] H0: P(category 3) = 0.1667 H0: P(category 4) = 0.1667
## [5] H0: P(category 5) = 0.1667 H0: P(category 6) = 0.1667
## [1] HA: The probabilities are different than those stated in H0
## [1] Degrees of freedom = 5
## [1] "------------------------------------------------------"
##            Observed Expected Distance
## category 1       15       20     1.25
## category 2       29       20     4.05
## category 3       16       20     0.80
## category 4       15       20     1.25
## category 5       30       20     5.00
## category 6       15       20     1.25
## [1] "------------------------------------------------------"
## [1] test statistic = 13.6
## [1] critical value = 11.0705
## [1] Pvalue = 0.0184
## [1] Decision: reject H0
## [1] ======================================================
```

(a) If the die is fair, what is the probability of each face? State the null and alternative hypotheses.

Each face has probability 1/6. H_0: the die is fair, $p_1 = p_2 = \cdots = p_6 = 1/6$. H_A: the die is not fair; at least one face has a probability different from 1/6.

(b) Fill in the expected frequency column and check the assumption that every expected count is at least 5.

Every expected count is $120 \times \frac{1}{6} = 20$, which is at least 5, so the assumption holds.

(c) Compute the test statistic χ^2_{obs} and give its degrees of freedom.

$$\chi^2_{obs} = \frac{(15-20)^2}{20} + \frac{(29-20)^2}{20} + \frac{(16-20)^2}{20} + \frac{(15-20)^2}{20} + \frac{(30-20)^2}{20} + \frac{(15-20)^2}{20}$$
$$= 1.25 + 4.05 + 0.80 + 1.25 + 5.00 + 1.25 = 13.6, \qquad df = 6 - 1 = 5$$

(d) Use the chi-square table to find the critical value and the p-value.

The critical value is $\chi^2_{5,0.95} = 11.070$. In the 5 df row, 13.6 falls between 12.833 (upper tail 0.025) and 16.750 (upper tail 0.005), so

$$0.005 < P(\chi^2 \geq 13.6 \mid H_0 \text{ true}) < 0.025$$

(the exact value is about 0.018).

(e) Is there enough evidence to conclude that the die is not fair?

Yes. The p-value $< 0.025 < \alpha = 0.05$ (equivalently, $\chi^2_{obs} = 13.6 > 11.070$), so we reject H_0. At the 5% significance level there is evidence that the die is not fair. The largest contributions come from the faces 2 and 5, which came up 29 and 30 times instead of the expected 20.

(**10**) Marital status of 400 young adults aged 18–24 compared with the U.S. population percentages, $\alpha = 0.05$.

```
chi.squared.GOF.test(observed.ct = c(140, 238, 2, 20), expected.freq = c(0.313, 0.561, 0.025, 0.101), al
```

```
## [1] ==================== test results ====================
## [1] H0: P(category 1) = 0.313 H0: P(category 2) = 0.561
## [3] H0: P(category 3) = 0.025 H0: P(category 4) = 0.101
## [1] HA: The probabilities are different than those stated in H0
## [1] Degrees of freedom = 3
## [1] "-------------------------------------------------------"
##            Observed Expected Distance
## category 1      140    125.2    1.750
## category 2      238    224.4    0.824
## category 3        2     10.0    6.400
## category 4       20     40.4   10.301
## [1] "-------------------------------------------------------"
## [1] test statistic = 19.2748
## [1] critical value = 7.8147
## [1] Pvalue = 2e-04
## [1] Decision: reject H0
## [1] =======================================================
```

Table 3: Expected counts and chi-square contributions for the marital status sample

Marital Status	Percent	Observed	Expected	Contribution
Never married	31.3	140	125.20	1.75
Married	56.1	238	224.40	0.82
Widowed	2.5	2	10.00	6.40
Divorced/separated	10.1	20	40.40	10.30

Assumptions. The 400 young adults are a random sample, and every expected count is at least 5 (the smallest is widowed, $400 \times 0.025 = 10$).

Hypotheses. H_0: the marital status of U.S. young adults aged 18–24 follows the population distribution, $p_{\text{never}} = 0.313$, $p_{\text{married}} = 0.561$, $p_{\text{widowed}} = 0.025$, $p_{\text{divorced}} = 0.101$. H_A: at least one of these proportions is different for young adults.

Test statistic. The expected counts are 400× each proportion: 125.20, 224.40, 10.00 and 40.40 (table above).

$$\chi^2_{obs} = \frac{(140-125.2)^2}{125.2} + \frac{(238-224.4)^2}{224.4} + \frac{(2-10)^2}{10} + \frac{(20-40.4)^2}{40.4} = 1.75+0.82+6.40+10.30 = 19.27, \qquad df = 4-1 = 3$$

***p*-value.** In the 3 df row, 19.27 is beyond 17.730 (upper tail 0.0005), so

$$P(\chi^2 \geq 19.27 \mid H_0 \text{ true}) < 0.0005$$

(the exact value is about 0.0002).

Decision. The p-value $< 0.0005 < \alpha = 0.05$ (equivalently, $\chi^2_{obs} = 19.27 > \chi^2_{3,0.95} = 7.815$), so we reject H_0. At the 5% significance level the marital status of young adults does not fit the distribution of the U.S. population. The biggest differences are that far fewer young adults are divorced/separated (20 vs 40.4 expected) or widowed (2 vs 10) than the population percentages predict.

Part VII: Chi-Square Test of Independence

(**11**) Sugar-sweetened beverage tax poll, 3×4 table, $n = 1{,}235$, $\alpha = 0.05$.

```r
tab = matrix(c(48, 433, 41, 160,
               54, 234, 24, 147,
               16,  43, 16,  19), nrow = 3, byrow = TRUE,
             dimnames = list(c('Against', 'In favor', 'No opinion'),
                             c('Asian-American', 'White/Non-Hispanic', 'Black', 'Hispanic/Latino')))
chi.squared.test(tab, var1.name = 'Ethnicity', var2.name = 'Opinion', alpha = 0.05, test = 'independenc
```

```
## [1] ==================== test results ====================
## [1] test type = independence
## [1] H0: The row variable and column variable are independent
## [1] HA: The row variable and column variable are dependent
## [1] Degrees of freedom = 6
## [1] -----------------------------------------------------
## [1] "Observed Counts"
##                              Ethnicity: category Asian-American
## Opinion: category Against                                    48
## Opinion: category In favor                                   54
## Opinion: category No opinion                                 16
## Column Total                                                118
##                              Ethnicity: category White/Non-Hispanic
## Opinion: category Against                                       433
## Opinion: category In favor                                      234
## Opinion: category No opinion                                     43
## Column Total                                                    710
##                              Ethnicity: category Black
## Opinion: category Against                           41
## Opinion: category In favor                          24
## Opinion: category No opinion                        16
## Column Total                                        81
##                              Ethnicity: category Hispanic/Latino Row Total
## Opinion: category Against                                    160       682
## Opinion: category In favor                                   147       459
## Opinion: category No opinion                                  19        94
## Column Total                                                 326      1235
## [1] "Expected Counts: (R * C)/n"
##                              Ethnicity: category Asian-American
## Opinion: category Against                                65.1628
## Opinion: category In favor                               43.8559
## Opinion: category No opinion                              8.9814
##                              Ethnicity: category White/Non-Hispanic
## Opinion: category Against                                  392.0810
## Opinion: category In favor                                 263.8785
## Opinion: category No opinion                                54.0405
##                              Ethnicity: category Black
## Opinion: category Against                      44.7304
## Opinion: category In favor                     30.1045
## Opinion: category No opinion                    6.1652
##                              Ethnicity: category Hispanic/Latino
## Opinion: category Against                                180.0259
## Opinion: category In favor                               121.1611
## Opinion: category No opinion                              24.8130
## [1] "Distances: (observed - expected)^2 / expected"
##                              Ethnicity: category Asian-American
## Opinion: category Against                                 4.5204
```

```
## Opinion: category In favor                                2.3464
## Opinion: category No opinion                               5.4848
##                              Ethnicity: category White/Non-Hispanic
## Opinion: category Against                                    4.2705
## Opinion: category In favor                                   3.3831
## Opinion: category No opinion                                 2.2556
##                              Ethnicity: category Black
## Opinion: category Against                     0.3111
## Opinion: category In favor                    1.2378
## Opinion: category No opinion                 15.6887
##                              Ethnicity: category Hispanic/Latino
## Opinion: category Against                                  2.2277
## Opinion: category In favor                                 5.5104
## Opinion: category No opinion                               1.3618
## [1] ------------------------------------------------------
## [1] test statistic = 48.5982
## [1] critical value = 12.5916
## [1] Pvalue = 0
## [1] Decision: reject H0
## [1] ======================================================
```

(a) State the null and alternative hypotheses in words.

H_0: opinion on the sugar-sweetened beverage tax is independent of ethnic group (the distribution of opinions is the same in every ethnic group). H_A: opinion on the tax depends on ethnic group.

(b) Compute the expected count for every cell and check the assumption that every expected count is at least 5.

Table 4: Expected counts: row total times column total, divided by 1,235

Opinion	Asian-American	White/Non-Hispanic	Black	Hispanic/Latino
Against tax	65.16	392.08	44.73	180.03
In favor of tax	43.86	263.88	30.10	121.16
No opinion	8.98	54.04	6.17	24.81

For example, the expected count for Asian-Americans against the tax is $\frac{682 \times 118}{1{,}235} = 65.16$. All twelve expected counts are in the table above; the smallest is 6.17 (Black, no opinion), so every expected count is at least 5 and the assumption holds.

(c) Compute the test statistic χ^2_{obs} and give its degrees of freedom.

Table 5: Each cell's contribution (O - E) squared over E

Opinion	Asian-American	White/Non-Hispanic	Black	Hispanic/Latino
Against tax	4.52	4.27	0.31	2.23
In favor of tax	2.35	3.38	1.24	5.51
No opinion	5.48	2.26	15.69	1.36

Adding the twelve contributions in the table above,

$$\chi^2_{obs} = \sum \frac{(O-E)^2}{E} = 48.60, \qquad df = (3-1)(4-1) = 6$$

(d) Use the chi-square table to find the critical value and the p-value.

The critical value is $\chi^2_{6,0.95} = 12.592$. In the 6 df row, 48.60 is far beyond 24.103 (upper tail 0.0005), so

$$P(\chi^2 \geq 48.60 \mid H_0 \text{ true}) < 0.0005$$

(the exact value is about 0.00000001).

(e) Is there enough evidence to conclude that opinion on the tax depends on ethnic group?

Yes. The p-value $< 0.0005 < \alpha = 0.05$ (equivalently, $\chi^2_{obs} = 48.60 > 12.592$), so we reject H_0. At the 5% significance level there is strong evidence that opinion on a sugar-sweetened beverage tax is related to ethnic group in California. The largest contribution is from Black respondents with no opinion (16 observed vs 6.17 expected); White/Non-Hispanic respondents were more often against the tax than expected (433 vs 392.08).